

# ALEX EFTIMIADIS

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## WORK EXPERIENCE

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### **Penguin Random House**

*Applied ML Scientist*

Remote

Nov 2022 - Present

- Built backward-looking MMM attribution models constrained to platform-reported ROI bounds, and forward-looking optimization models projecting ROI on hypothetical spend scenarios.
- Designed constrained neural net architecture (JAX) encoding spend-response shape and monotonicity priors to correct for platform attribution bias; used xgboost where data supported it.
- Models projected to save \$4M/year via spend constraints and allocation.
- Presented modeling approach at Data Science Salon (2023) and ANA Marketing Conference (2025).
- Applied difference-in-differences to estimate incremental lift on channels with limited spend data.
- Built Claude Code pre-tool-use hook intercepting error-swallowing patterns (silent excepts, broad fallbacks, silent default injections) using 12 regex classifiers and a three-tier decision system — clean pass, hard deny, or LLM judge; blocks 2 violations per session (avg) and adopted across the team.
- Built autonomous error-resolution agent (Linux daemon) that monitors production pipeline failures via Slack, files merge requests using Claude Code, and verifies fixes by rerunning failing code and checking for traceback absence logs and changes in prod data.
- Supervised an intern data scientist.
- Built and deployed AI driven ad generation and monitoring pipeline using Python and Kubernetes. Ads deployed to Facebook, Instagram, and Google.
- Analyzed campaign coverage and efficacy across products and platforms using Python and SQL.

### **SPAR AI**

*Research Mentee*

Remote

Sep 2026 - Present

- Investigating whether topological features of LLM activations improve deception detection over linear probes, beginning with replication and benchmarking on established deception scenarios.

### **Trekkit LLC**

*Founder*

Remote

Jan 2024 - Present

- Built and launched several financial planning apps on decisiondough.com (SST/AWS).
- Built and launched Offensive Weather to Google Play Store and Apple App Store (SST/AWS).

### **FINRA**

*Lead Data Scientist*

Remote

June 2019 - Nov 2022

- Led deployment of NLP models in production (Docker, Lambda/AWS), reducing costs by 80%.
- Led R&D and open sourced the Model Validation Toolkit — a research library implementing constrained interpretable neural nets, Bayesian credibility analysis, and divergence-based drift detection.
- Developed supervised and unsupervised models to identify insider trading (xgboost; 96% AUC), market manipulation (dbscan), fraud (bayesian analysis), and triage external communication (xgboost, sklearn, BERT).
- Mentored junior data scientists and led regular data science sessions and workshops.
- Gave internal talks on: software engineering for data scientists, countering sample bias, measuring model drift, thresholding, normalizing flows.
- Developed and conducted technical interview process and brought on 7 data scientists.
- Promoted twice in three years.

### **Deepsig**

*Machine Learning Engineer*

Arlington, VA

January 2019 - March 2019

- Designed and implemented deep learning based signal detector and classifier.
- Compared deep learning approaches benchmarked against classical clustering algorithms for signal identification and classification.

- Gave talk on semi-supervised learning.

**Catalist LLC**  
*Analytics Engineer*

Washington DC  
February 2018 - January 2019

- Optimized, parallelized, and deployed NLP Keras model.
- Wrote SQL parser that refactored over one million lines of legacy SQL scripts.
- Designed and wrote data processing pipeline for election results as they became available the night of the election.
- Wrote internal technical guides on parallel processing.
- Contributed code to Keras (fixed tokenizer).

**Comsol**  
*Developer*

Burlington, MA  
February 2016 - May 2017

- Researched models and techniques to simulate physical phenomena of interest to engineers and scientists.
- Implemented and documented algorithms used for numerical simulations and user interfaces in Java.
- Helped customers create and optimize simulations.

## SKILLS

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|------------------------|--------------------------------------------------------|
| Programming Languages: | Python, Bash, SQL                                      |
| Frameworks:            | Jax/Pytorch, Numpy/Scipy, Cython, Pandas, Scikit-learn |
| Tools:                 | Git, Vim, AWS, Kubernetes, Jupyter, Plotly, Docker     |

## PROJECTS

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**Fitness Mortality Calculator (Apr 2026)** *Python* <https://aefitima.github.io/fitness-mortality/>  
Evidence-based web app modeling all-cause mortality risk from VO2 max and grip strength; anchors to SSA life tables using published hazard ratios, normative percentile splines, and population-mean normalization via Gauss-Legendre quadrature.

**Tlang (March 2022)** *Python* <https://github.com/aefitima/tlang>  
Production-grade DSL for composing context-sensitive transpilers with template-based code generation.

**Model Validation Toolkit (Dec 2021)** *Python* <https://github.com/FINRAOS/model-validation-toolkit>  
Research-grade open source library for ML validation and monitoring, featuring original work on constrained interpretable neural networks, Bayesian small-sample credibility analysis, information-theoretic drift detection (KL, JS, Wasserstein divergences), adaptive thresholding, bias-variance decomposition, and Sobol sensitivity analysis.

**Discrete Exterior Calculus Framework (May 2014)** *Python, Cython, CUDA*  
<https://github.com/aefitima/kahler>  
GPU-parallelized discrete exterior calculus framework for Hermitian manifolds; extends PyDEC with Cython/CUDA and supports arbitrary metrics, periodic boundary conditions, and cohomological operators.

**Hexchat (July 2013)** *Python* <https://github.com/aefitima/hexchat>  
Custom TCP-over-XMPP tunneling protocol for Tor, supporting multi-account JID rotation and ordered delivery across arbitrary chat relays to bypass censorship bandwidth limits.

## AWARDS AND PUBLICATIONS

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**Leveraging Data Science to Optimize Retail Media Campaigns** ANA Marketing Conference  
Presented research on mixed market models at ANA Marketing Conference. Nov 2025

**Advertising at Scale** Data Science Salon  
Presented work on automated advertising and A/B testing at Data Science Salon. June 2023

**Introducing the Model Validation Toolkit** Open Source Data Science (ODSC) East 2022  
Gave talk at ODSC East introducing open source toolkit developed internally at FINRA. April 2022

**Enhancing the Three-Dimensional Structure of Adherent Gingival Fibroblasts and Spheroids via a Fibrous Protein-Based Hydrogel Cover** Cells Tissues Organs  
Published with biologists at the American Dental Association Foundation. August 2016

**Kahler: An Implementation of Discrete Exterior Calculus on Hermitian Manifolds**  
arXiv:1405.7879

Independent research and implementation of a GPU-parallelized finite elements framework for Hermitian manifolds. May 2014

EDUCATION

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| <b>UMBC</b>                       | Catonsville, MD |
| BS Physics (Minor in Mathematics) | 2013 - 2015     |